



Digital Egypt Pioneers Initiative Fortinet Project Implementing VPN Solutions with FortiGate

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Agenda

- **Introduction**
- **Week 1 Deliverables**
- **Week 2 Deliverables**
- **Week 3 Deliverables**
- **Conclusion**
- **Q&A**

Overview of VPNs and Their Importance

A Virtual Private Network (VPN) is a secure connection that allows users to access a private network over the internet. VPNs encrypt data traffic, ensuring confidentiality, integrity, and anonymity. They are crucial for businesses and individuals as they:

- Protect sensitive data from cyber threats.
- Enable secure remote access for employees.
- Ensure privacy when using public networks.
- Facilitate safe communication between branch offices or across geographies.

- A leading global provider of cybersecurity and networking solutions
- Established in 2000, trusted by enterprises, governments, and service providers
- Creator of FortiGate, one of the most widely deployed next-generation firewalls

Solution Areas

- Network security (NGFW, IPS, Web Filtering)
- Secure remote access & VPN technologies
- SD-WAN and branch connectivity
- Cloud and data center protection



Why FortiGate for VPN Solutions

FortiGate, developed by Fortinet, is a leading network security platform that offers robust VPN solutions. It is widely chosen because it:

- Provides high-performance, hardware-accelerated VPN connections.
- Supports both SSL and IPsec VPNs for flexible deployment.
- Integrates advanced security features like firewall, intrusion prevention, and malware protection.
- Offers easy management through a centralized interface, making VPN deployment and monitoring efficient.



Project Stages

1- SSL VPN Configuration

- Remote access for users
- Authentication, portal settings, and access policies

2- IPsec Site-to-Site VPN

- Private communication between two networks
- Routing and traffic selectors to enable encrypted connectivity

3- SD-WAN Integration

- Link performance monitoring
- Intelligent VPN traffic steering for reliability and continuity

Week 1 – VPN Concepts & SSL VPN Configuration on FortiGate

What is a VPN?

A Virtual Private Network (VPN) is a technology that creates a secure, encrypted tunnel between a user and a network over the internet.

Why do we use a VPN?

- Protects data from interception
- Allows safe remote access to internal resources
- Connects networks securely over public channels
- Ensures privacy and confidentiality

Types of VPNs

Site-to-Site VPN (Network ↔ Network)

- Connects entire networks (e.g., HQ ↔ branch) .
- Used for permanent connections .
- Ideal for businesses with multiple locations .
- Runs mainly on IPsec .
- Always-on and transparent to users .

Remote Access VPN (User ↔ Network)

- Allows individual users to connect remotely .
- Ideal for employees working from home/outside .

Two main types :

SSL VPN

IPsec Remote Access

Requires user authentication .

SSL VPN vs IPsec VPN

• **SSL VPN**

Works over HTTPS (TCP 443)

Uses SSL/TLS encryption

Easy to access (browser/FortiClient)

Works smoothly behind NAT & firewalls

Best for remote users / work from home

• **IPsec VPN**

Uses IPsec protocols .

Requires a VPN client .

More secure for managed corporate devices .

Very stable for site-to-site tunnels .

Preferred for permanent office connections .

Configuration (Week 1 – SSL VPN)

Step 1 – Interface Configuration:

- Port1 (WAN) was configured to receive a DHCP IP address for external connectivity.**
- Port2 (LAN) was assigned the static subnet 192.168.100.1/24 to serve internal hosts.**

Interfaces were verified to ensure proper WAN and LAN separation.

FortiGate-VM64

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface Search

Group By Type

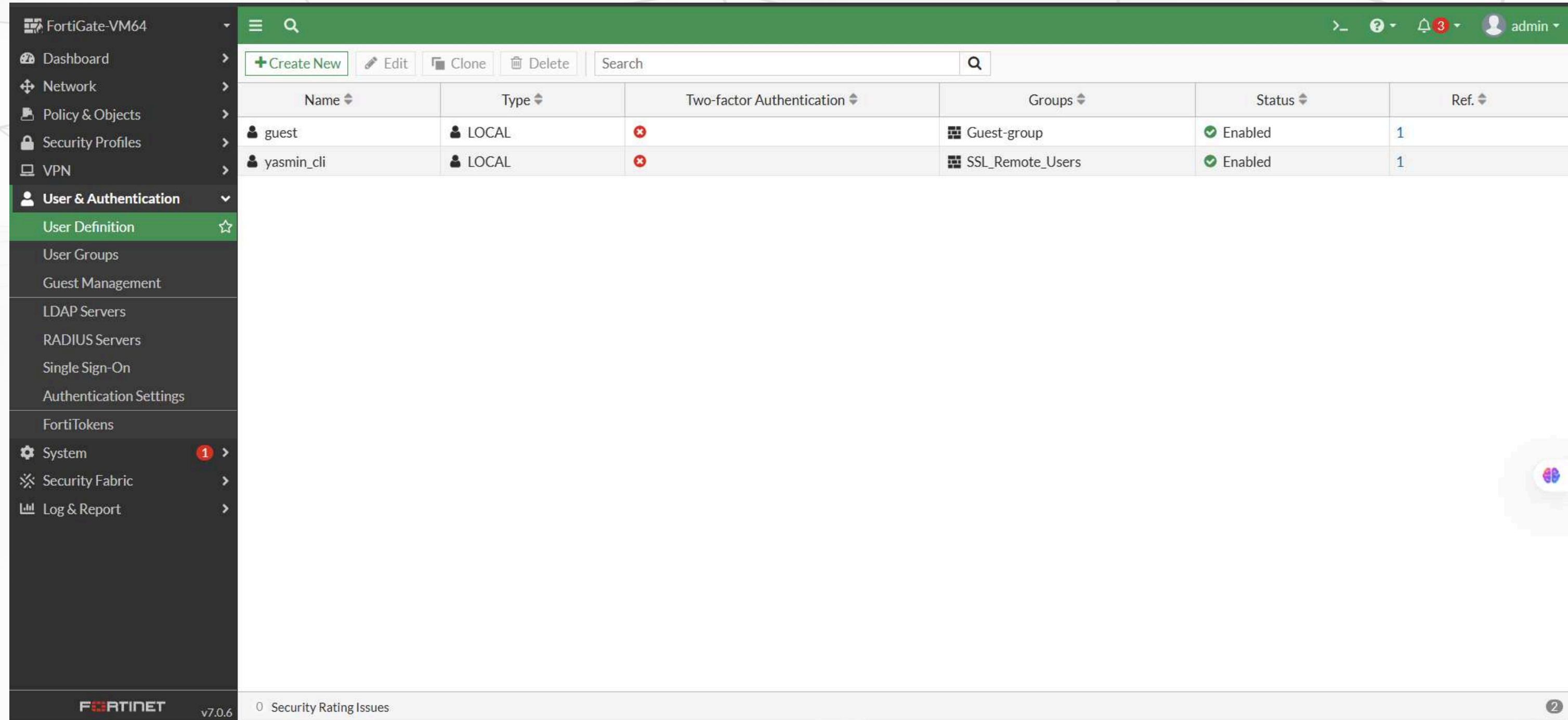
Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2
Physical Interface 10							
port1	Physical Interface		192.168.198.132/255.255.255.0	PING HTTPS SSH HTTP FMG-Access			1
port2	Physical Interface		192.168.100.1/255.255.255.0	PING HTTPS HTTP	1	192.168.100.50-192.168.100.200	4
port3	Physical Interface		0.0.0.0/0.0.0.0				0
port4	Physical Interface		0.0.0.0/0.0.0.0				0
port5	Physical Interface		0.0.0.0/0.0.0.0				0
port6	Physical Interface		0.0.0.0/0.0.0.0				0
port7	Physical Interface		0.0.0.0/0.0.0.0				0
port8	Physical Interface		0.0.0.0/0.0.0.0				0

0 Security Rating Issues

0% 12 Updated: 09:12:36

Step 2 – Local User Creation:

- A local user (yasmin_cli) was created for SSL-VPN authentication.
- The user was assigned a secure password and enabled for login.



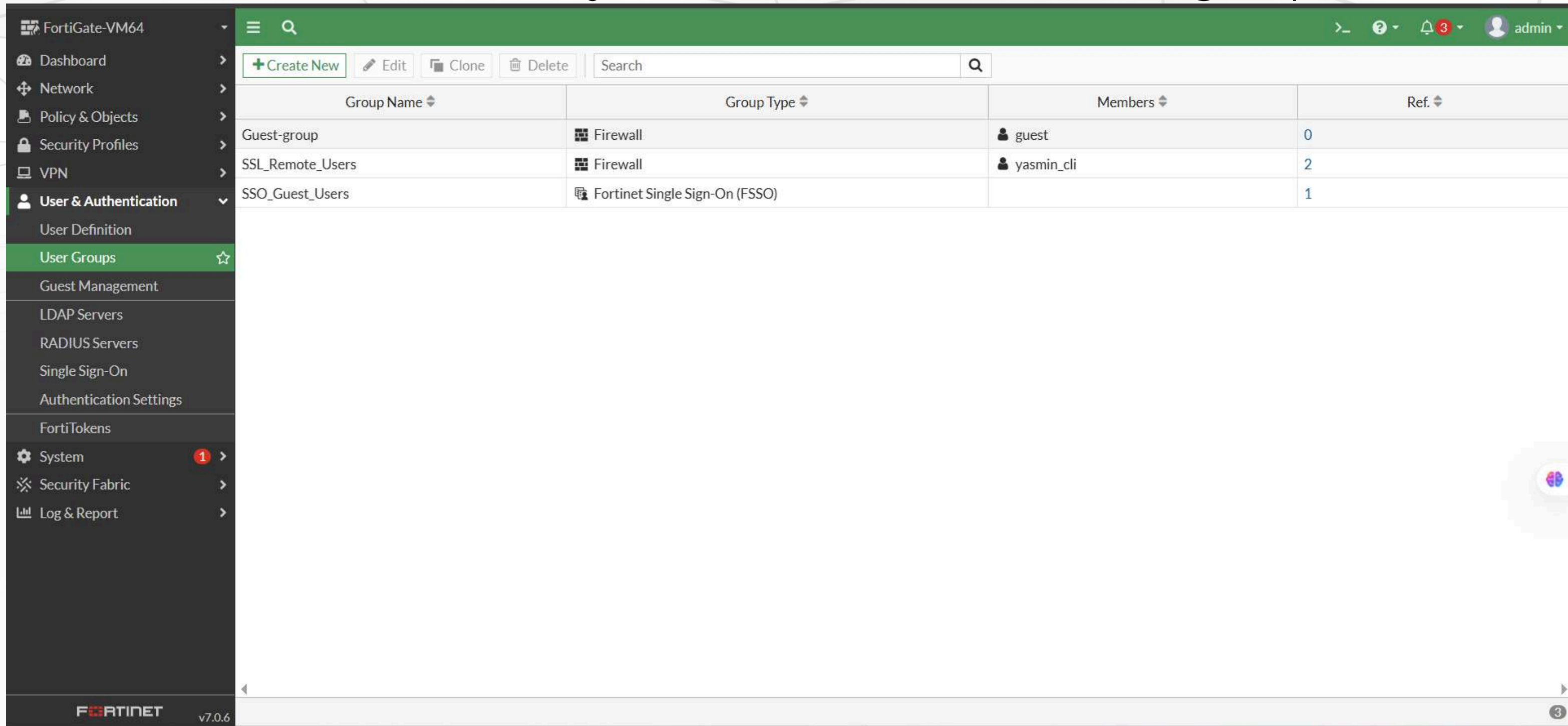
The screenshot shows the FortiGate VM64 User & Authentication interface. The left sidebar contains the navigation menu with 'User & Authentication' expanded. The main content area displays a table of user definitions.

Name	Type	Two-factor Authentication	Groups	Status	Ref.
guest	LOCAL	✗	Guest-group	✓ Enabled	1
yasmin_cli	LOCAL	✗	SSL_Remote_Users	✓ Enabled	1

The bottom status bar indicates '0 Security Rating Issues' and the version 'v7.0.6'.

Step 3 – User Group Configuration:

- A dedicated group named SSL_Remote_Users was created.
- The local user yasmin_cli was added to this group.



The screenshot shows the FortiGate VM64 web interface. The left sidebar contains a navigation menu with the following items: Dashboard, Network, Policy & Objects, Security Profiles, VPN, User & Authentication (expanded), User Definition, User Groups (highlighted with a star), Guest Management, LDAP Servers, RADIUS Servers, Single Sign-On, Authentication Settings, FortiTokens, System (with a red notification badge), Security Fabric, and Log & Report. The main content area displays the 'User Groups' configuration page. At the top, there are buttons for '+ Create New', 'Edit', 'Clone', and 'Delete', along with a search bar. Below this is a table with the following columns: Group Name, Group Type, Members, and Ref. The table contains three entries:

Group Name	Group Type	Members	Ref.
Guest-group	Firewall	guest	0
SSL_Remote_Users	Firewall	yasmin_cli	2
SSO_Guest_Users	Fortinet Single Sign-On (FSSO)		1

The bottom of the interface shows the Fortinet logo and version v7.0.6. A small circular icon with a brain symbol is visible in the bottom right corner of the main content area.

Step 4 – SSL VPN Settings:

- SSL-VPN was enabled on port1.



FortiGate-VM64

SSL-VPN Settings

Enable SSL-VPN ☒

Listen on Interface(s) port1

Listen on Port 10443

Web mode access will be listening at <https://192.168.198.132:10443>

Server Certificate Fortinet_Factory

You are using a default built-in certificate, which will not be able to verify your server's domain name (your users will see a warning). Let's Encrypt can be used to easily generate a trusted certificate if you do not have one.

Create Certificate

Redirect HTTP to SSL-VPN ☐

Restrict Access Allow access from any host Limit access to specific hosts

Idle Logout ☒

Inactive For 300 Seconds

Require Client Certificate ☐

Tunnel Mode Client Settings

Address Range Automatically assign addresses Specify custom IP ranges

Tunnel users will receive IPs in the range of 10.212.134.200 - 10.212.134.210

DNS Server Same as client system DNS Specify

Specify WINS Servers ☐

Authentication/Portal Mapping

Create New Edit Delete Send SSL-VPN Configuration

Users/Groups	Portal
SSL_Remote_Users	Remote_User_Portal
All Other Users/Groups	Remote_User_Portal

Additional Information

API Preview

Edit in CLI

SSL VPN Setup Guides

Web Mode

Web Mode for Remote User

Tunnel Mode

Full Tunnel for Remote User

Split Tunnel for Remote User

Tunnel Mode Host Check

Multi-realm

Multi-Realm

Authentication

Certificate Authentication

LDAP-Integrated Certificate Authentication

FortiToken Mobile Push Authentication

RADIUS on FortiAuthenticator

RADIUS and FortiToken Mobile Push on FortiAuthenticator

Local User Password Policy

RADIUS Password Renew on FortiAuthenticator

LDAP User Password Renew

VPN Setup on FortiClient

Configuring an SSL VPN Connection

Troubleshooting

Troubleshooting

Documentation

Online Help

Video Tutorials

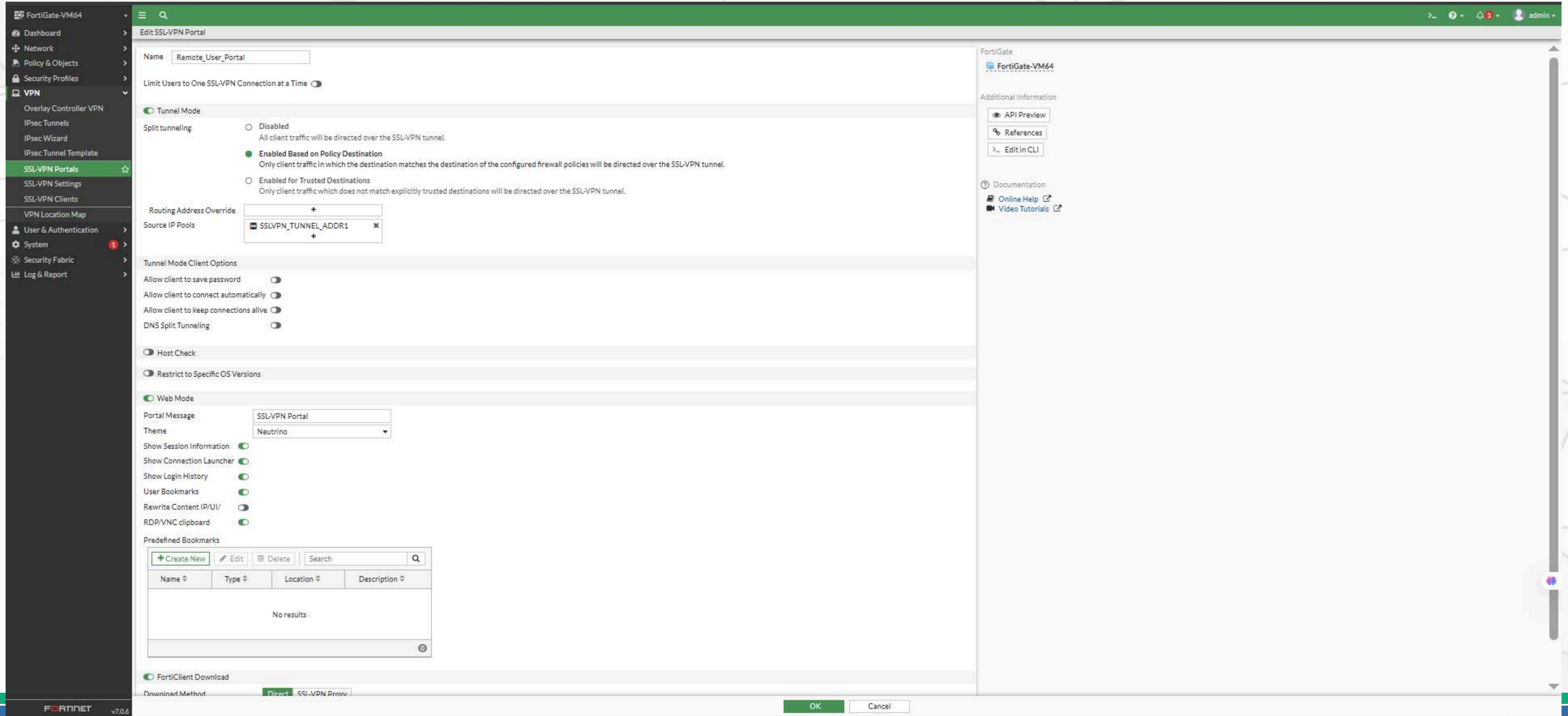
Security Rating Issues

Show Dismissed ☐

Apply

Step 5 – SSL-VPN Portal Setup:

- The Remote_User_Portal was configured.
- Split Tunneling was enabled to route internal traffic only.
- Routing override was set to the SSLVPN address pool.



The screenshot shows the FortiGate VM64 configuration interface for the SSL-VPN Portal. The left sidebar contains navigation options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller VPN, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals (selected), SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report.

The main configuration area is titled "Edit SSL-VPN Portal" and shows the "Remote_User_Portal" configuration. The "Tunnel Mode" section is active, with "Split tunneling" set to "Enabled Based on Policy Destination". The "Routing Address Override" is set to "SSLVPN_TUNNEL_ADDR1". The "Web Mode" section is also active, with the "Portal Message" set to "SSL-VPN Portal" and the "Theme" set to "Neutrino".

The "Predefined Bookmarks" section shows a table with columns: Name, Type, Location, and Description. The table is currently empty, displaying "No results".

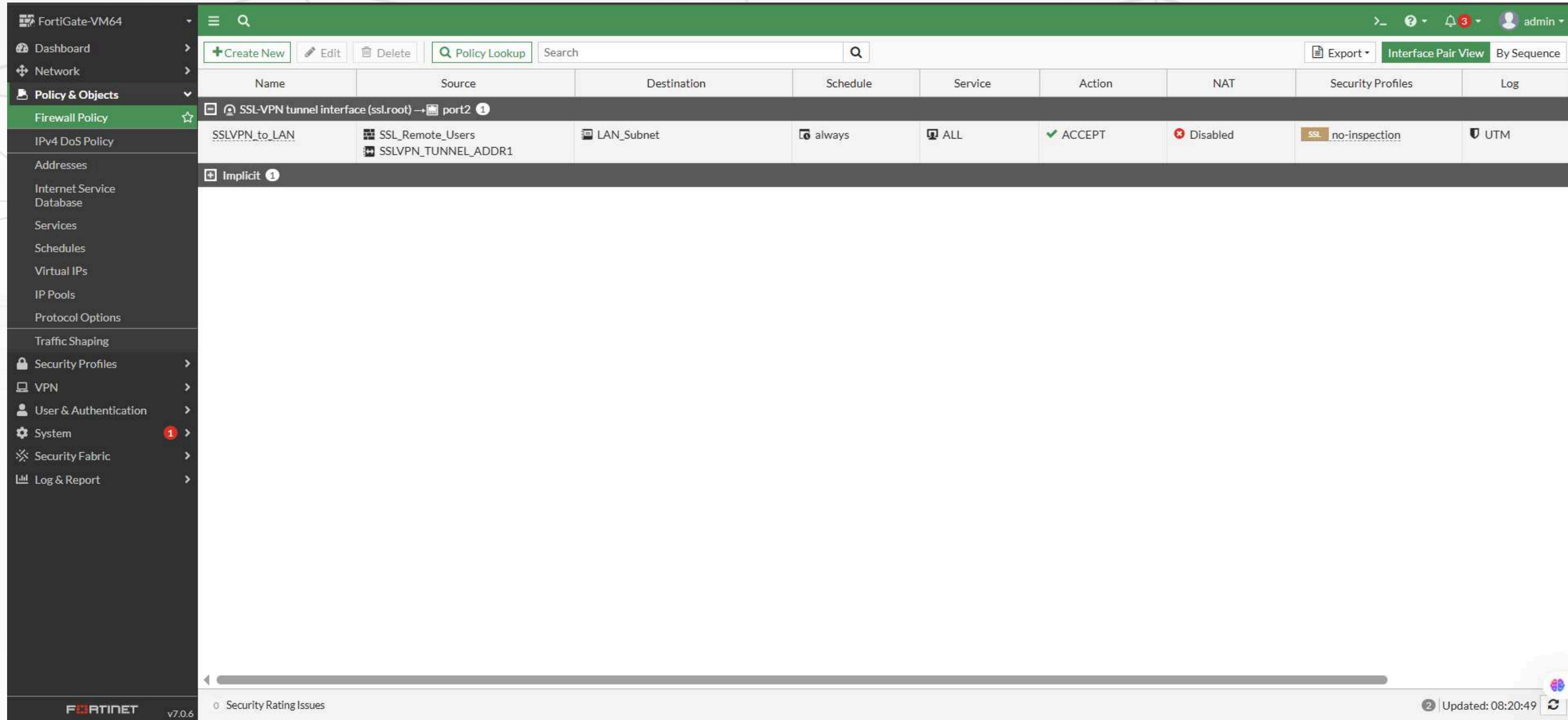
The "FortiClient Download" section is visible at the bottom, with the "Download Method" set to "Direct".

The right sidebar contains "Additional Information" links: API Preview, References, and Edit in CLI. It also includes "Documentation" links: Online Help and Video Tutorials.

The bottom of the interface shows the FortiGate logo and version v7.0.6, along with "OK" and "Cancel" buttons.

Step 6 – Firewall Policies:

- Policy created to allow SSLVPN → LAN user access to internal resources.



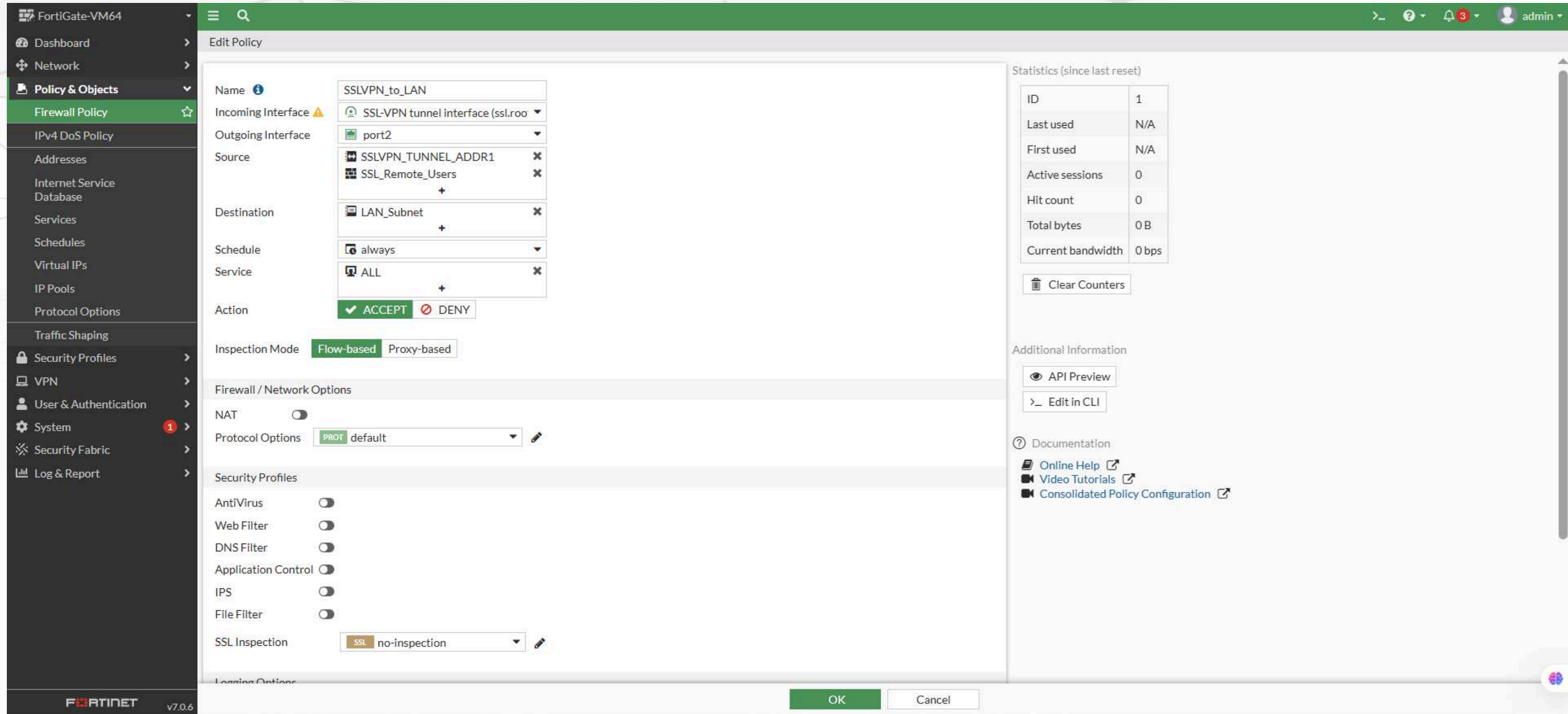
The screenshot shows the FortiGate-VM64 web interface for configuring Firewall Policies. The left sidebar contains the navigation menu with 'Policy & Objects' expanded and 'Firewall Policy' selected. The main area displays a table of firewall policies.

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log
SSLVPN to LAN	SSL_Remote_Users SSLVPN_TUNNEL_ADDR1	LAN_Subnet	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM
Implicit								

The bottom status bar shows 'Security Rating Issues' and 'Updated: 08:20:49'.

Step 6 – Firewall Policies:

- Policy created to allow SSLVPN → LAN user access to internal resources.



The screenshot shows the FortiGate-VM64 web interface for editing a Firewall Policy named "SSLVPN_to_LAN". The configuration is as follows:

- Name:** SSLVPN_to_LAN
- Incoming Interface:** SSL-VPN tunnel interface (ssl.root)
- Outgoing Interface:** port2
- Source:** SSLVPN_TUNNEL_ADDR1, SSL_Remote_Users
- Destination:** LAN_Subnet
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT (checked), DENY (unchecked)
- Inspection Mode:** Flow-based (selected), Proxy-based (unselected)
- Firewall / Network Options:**
 - NAT:** disabled
 - Protocol Options:** default
- Security Profiles:**
 - AntiVirus: disabled
 - Web Filter: disabled
 - DNS Filter: disabled
 - Application Control: disabled
 - IPS: disabled
 - File Filter: disabled
 - SSL Inspection: no-inspection

Statistics (since last reset):

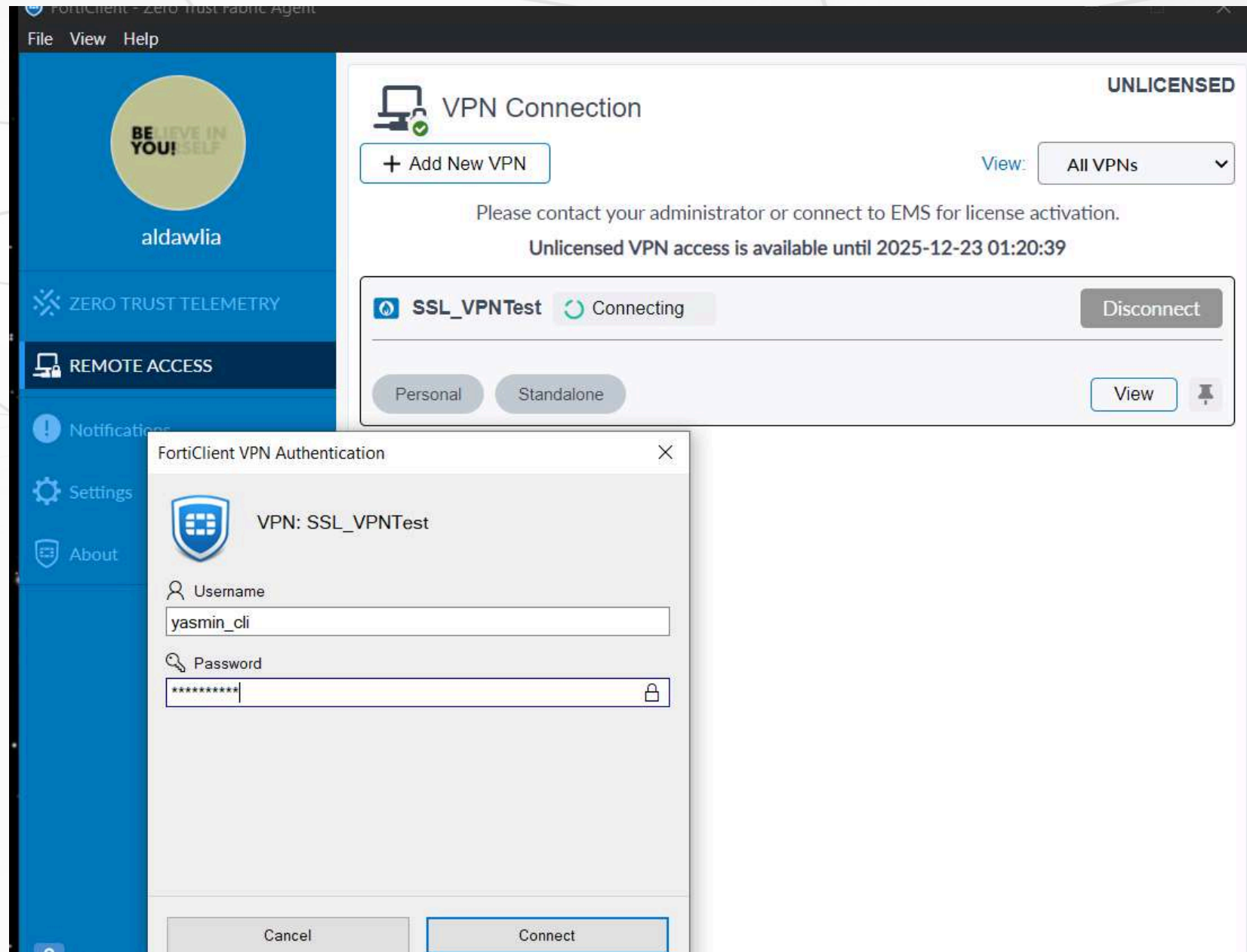
ID	1
Last used	N/A
First used	N/A
Active sessions	0
Hit count	0
Total bytes	0 B
Current bandwidth	0 bps

Additional Information:

- API Preview
- Edit in CLI
- Documentation:
 - Online Help
 - Video Tutorials
 - Consolidated Policy Configuration

At the bottom, there are "OK" and "Cancel" buttons.

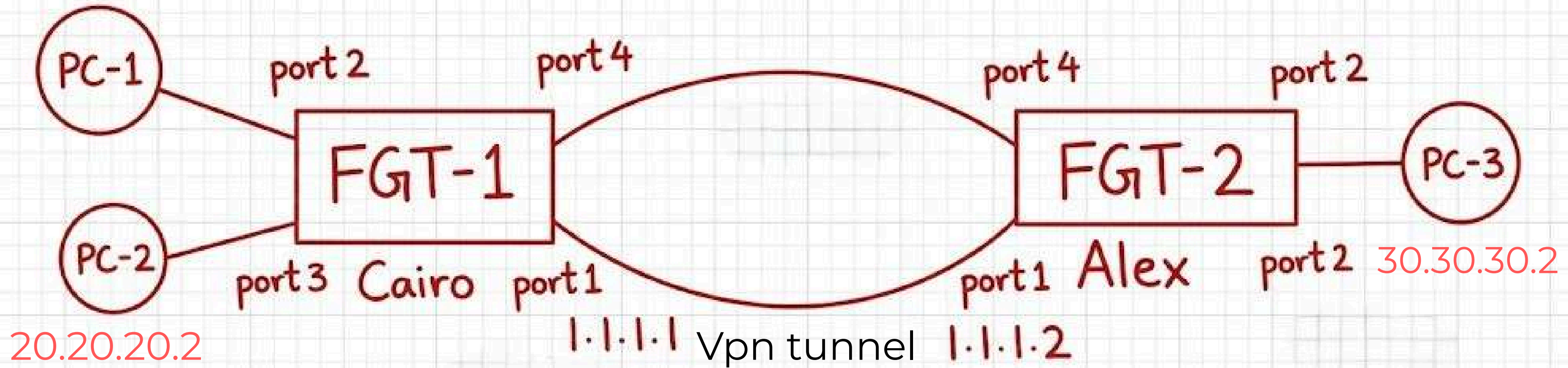
TEST



Week2: IPsec Site-to-Site VPN Tunnel Implementation

Topology Overview

10.10.10.2



Goal Traffic:

Secure communication primarily between the 10.10.10.0/24 and 30.30.30.0/24 networks.

Configuration

Step 1 – Interface Configuration:

- Port1 was configured as the management interface on both firewalls.
- IP addresses were assigned to all required interfaces to provide both internal and external connectivity.
- Each site was configured with its correct internal subnets and WAN interfaces.

Cairo

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
802.3ad Aggregate						
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254
Physical Interface						
LAN 1 (port2)	Physical Interface		10.10.10.1/255.255.255.0	PING	1	10.10.10.2-10.10.10.254
LAN 2 (port3)	Physical Interface		20.20.20.1/255.255.255.0	PING	1	20.20.20.2-20.20.20.254
port1	Physical Interface		192.168.1.13/255.255.255.0	PING HTTPS SSH HTTP FMG-Access		
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			

Security Rating Issues

0% Updated: 05:42:56

Alex

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
802.3ad Aggregate						
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254
Physical Interface						
LAN (port2)	Physical Interface		30.30.30.1/255.255.255.0	PING	1	30.30.30.2-30.30.30.254
port1	Physical Interface		192.168.1.11/255.255.255.0	PING HTTPS SSH HTTP FMG-Access		
port3	Physical Interface		0.0.0.0/0.0.0.0			
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			

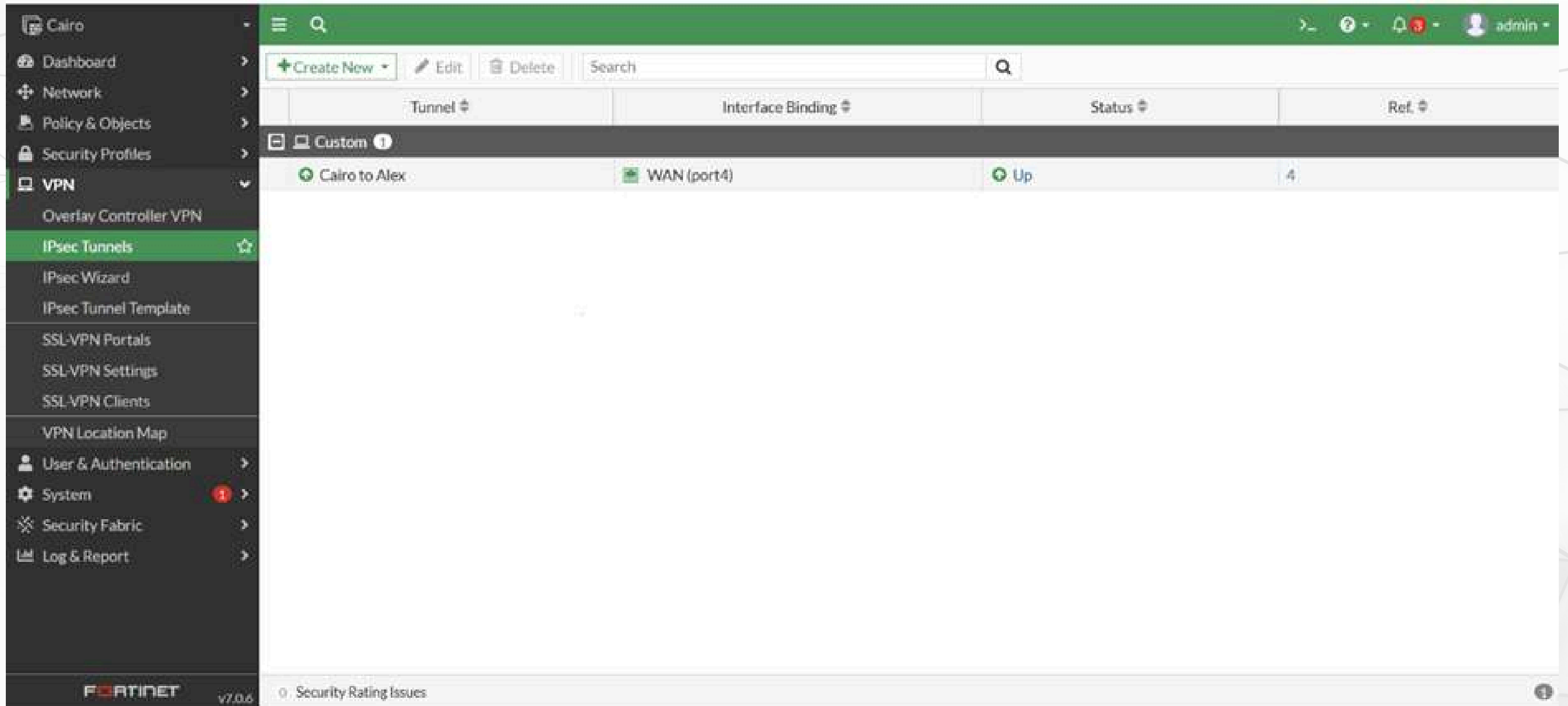
Security Rating Issues

0% Updated: 05:51:31

Configuration

Step 2 – VPN Tunnel Configuration

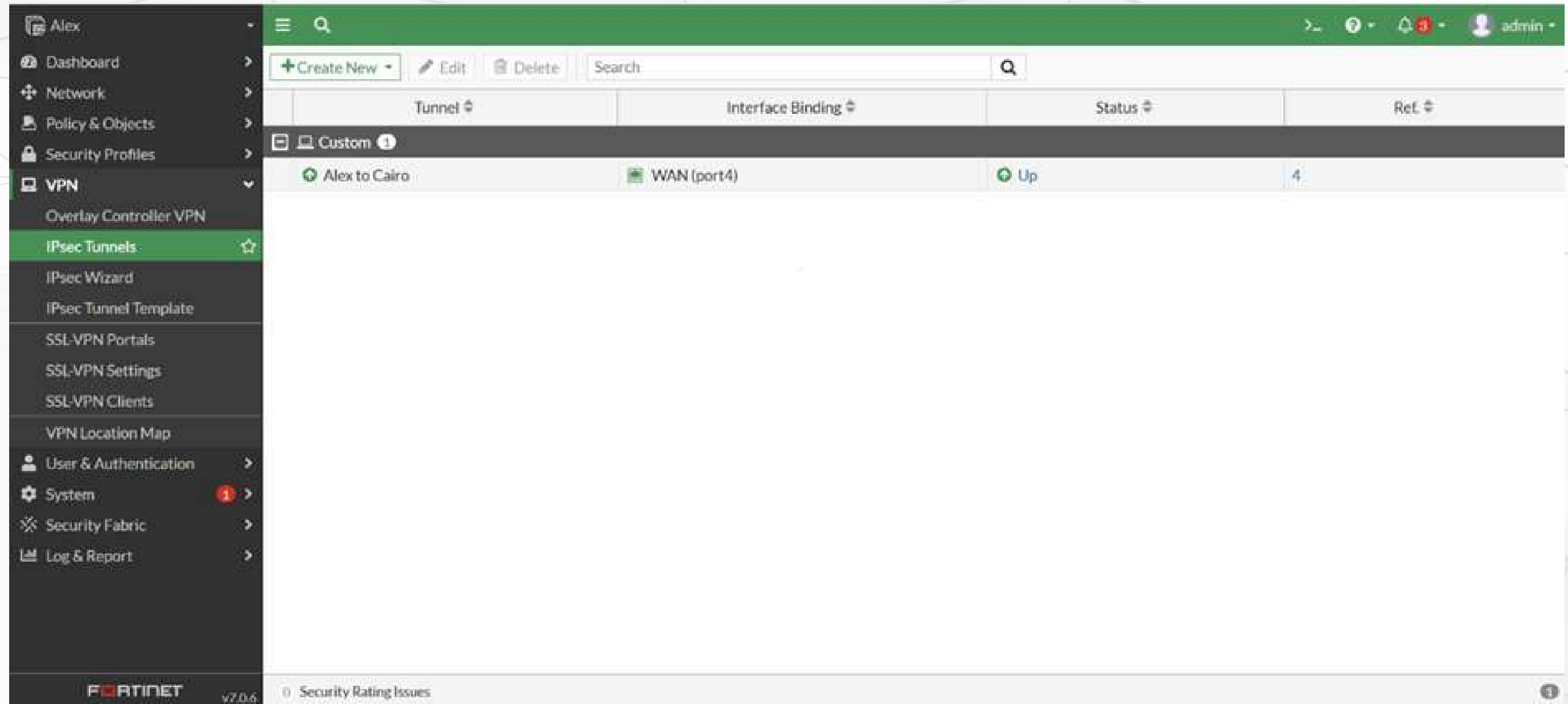
- An IPsec VPN tunnel was created on both firewalls.
- The traffic selectors were configured as follows:
 - Cairo: 10.10.10.0/24 → 30.30.30.0/24
 - Alex: 30.30.30.0/24 → 10.10.10.0/24
- Phase 1 and Phase 2 parameters were matched to ensure successful negotiation of the VPN tunnel.



The screenshot displays the Fortinet FortiGate v7.0.6 web interface for configuring VPNs. The left sidebar contains a navigation menu with the following items: Cairo, Dashboard, Network, Policy & Objects, Security Profiles, VPN (expanded), Overlay Controller VPN, IPsec Tunnels (highlighted), IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System (with a red notification badge), Security Fabric, and Log & Report. The main content area shows the 'IPsec Tunnels' configuration page. At the top, there is a green header bar with a search icon and a user profile 'admin'. Below this is a toolbar with 'Create New', 'Edit', 'Delete', and a search field. The main table lists the configured tunnels:

Tunnel	Interface Binding	Status	Ref.
Custom 1			
Cairo to Alex	WAN (port4)	Up	4

The bottom status bar shows '0 Security Rating Issues' and the FortiGate logo with version 'v7.0.6'.



The screenshot displays the Fortinet FortiGate web interface. On the left is a dark sidebar menu with the following items: Alex, Dashboard, Network, Policy & Objects, Security Profiles, VPN (expanded), Overlay Controller VPN, IPsec Tunnels (highlighted), IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System (with a red notification icon), Security Fabric, and Log & Report. The main content area has a green header bar with a search icon and a user profile 'admin'. Below the header is a toolbar with '+ Create New', 'Edit', 'Delete', and a search bar. A table lists the IPsec tunnels:

Tunnel	Interface Binding	Status	Ref.
Custom ⓘ			
Alex to Cairo	WAN (port4)	Up	4

At the bottom, the FortiGate logo and version 'v7.0.6' are on the left, and '0 Security Rating Issues' with a notification icon are on the right.

Configuration

Step 3 – Static Route Configuration

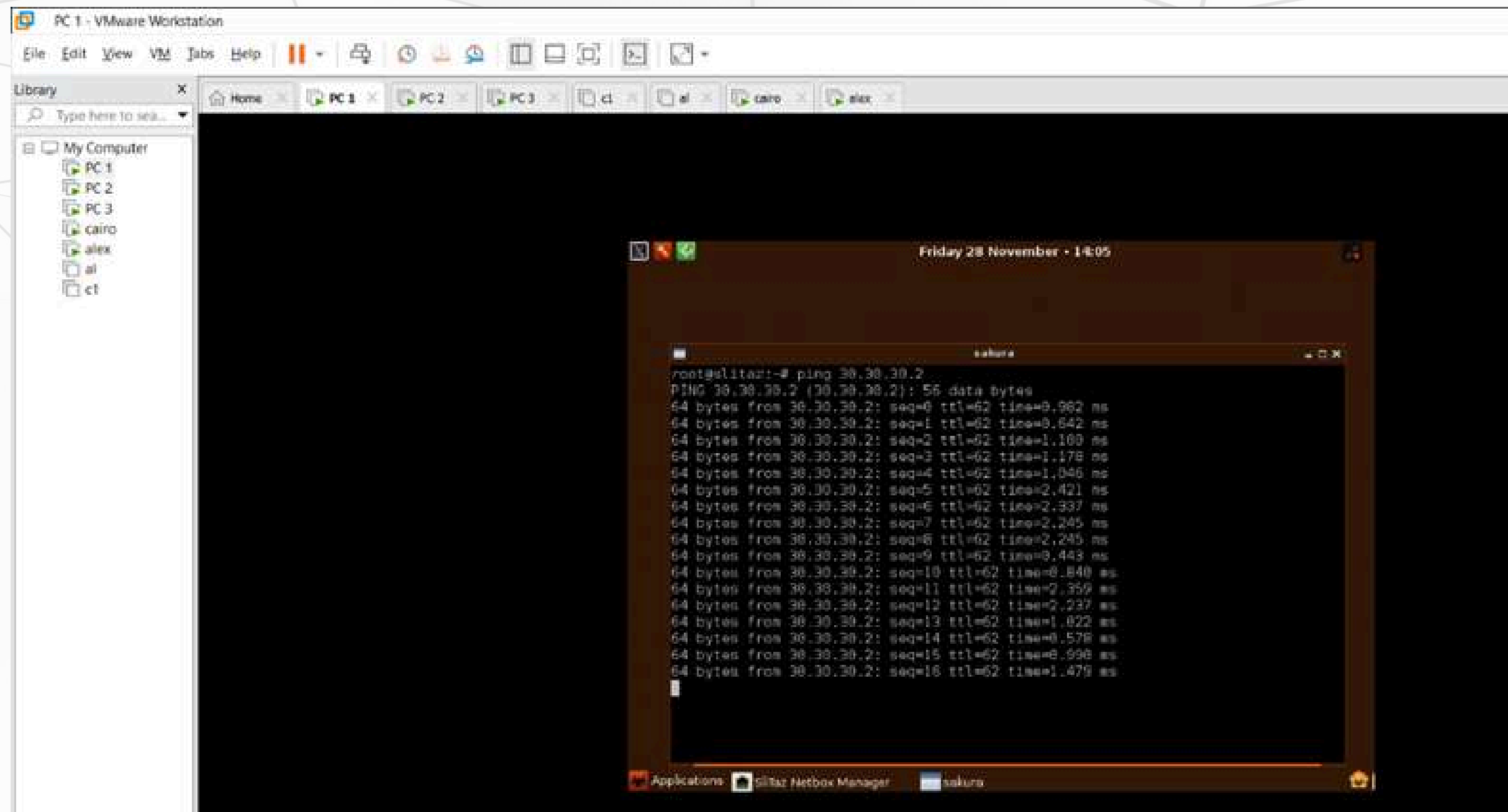
- Static routes were added on each firewall so that traffic destined for the remote network is forwarded into the VPN tunnel.
- This ensures correct routing between the Cairo and Alex networks.

Step 4 – Firewall Policy Configuration

- Security policies were created to allow traffic between the internal networks and the VPN tunnel interface.
- Only traffic that matches the defined traffic selector is allowed through the tunnel.

Connectivity Test Results

Ping Test: PC-1 to PC-3

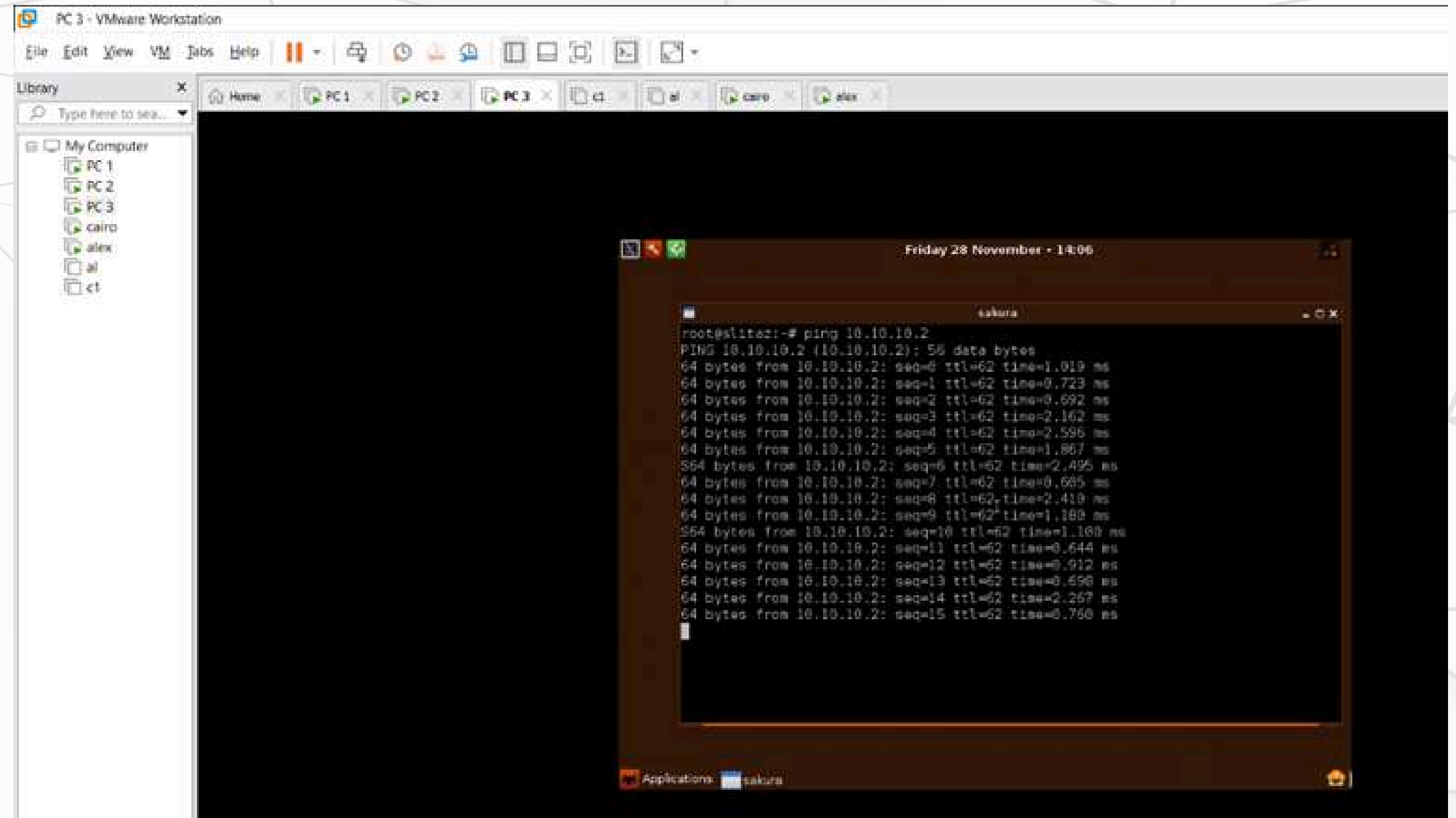


The screenshot shows a VMware Workstation interface with a terminal window open. The terminal displays the results of a ping test from PC-1 to PC-3. The test shows 16 successful pings with varying response times.

```
root@slitaz:~# ping 30.30.30.2
PING 30.30.30.2 (30.30.30.2): 56 data bytes
64 bytes from 30.30.30.2: seq=0 ttl=62 time=0.962 ms
64 bytes from 30.30.30.2: seq=1 ttl=62 time=0.642 ms
64 bytes from 30.30.30.2: seq=2 ttl=62 time=1.100 ms
64 bytes from 30.30.30.2: seq=3 ttl=62 time=1.178 ms
64 bytes from 30.30.30.2: seq=4 ttl=62 time=1.046 ms
64 bytes from 30.30.30.2: seq=5 ttl=62 time=2.421 ms
64 bytes from 30.30.30.2: seq=6 ttl=62 time=2.337 ms
64 bytes from 30.30.30.2: seq=7 ttl=62 time=2.345 ms
64 bytes from 30.30.30.2: seq=8 ttl=62 time=2.245 ms
64 bytes from 30.30.30.2: seq=9 ttl=62 time=0.443 ms
64 bytes from 30.30.30.2: seq=10 ttl=62 time=0.840 ms
64 bytes from 30.30.30.2: seq=11 ttl=62 time=2.359 ms
64 bytes from 30.30.30.2: seq=12 ttl=62 time=2.237 ms
64 bytes from 30.30.30.2: seq=13 ttl=62 time=1.022 ms
64 bytes from 30.30.30.2: seq=14 ttl=62 time=0.576 ms
64 bytes from 30.30.30.2: seq=15 ttl=62 time=0.996 ms
64 bytes from 30.30.30.2: seq=16 ttl=62 time=1.479 ms
```

Connectivity Test Results

Ping Test: PC-3 to PC-1



The screenshot shows a VMware Workstation window titled "PC 3 - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a Library pane on the left. The Library pane shows a tree structure under "My Computer" with items: PC 1, PC 2, PC 3, cairo, alex, al, and ct. The main window displays a terminal window titled "sakura" with the following output:

```
Friday 28 November - 14:06
root@slite2:~# ping 10.10.10.2
PING 10.10.10.2 (10.10.10.2): 56 data bytes
64 bytes from 10.10.10.2: seq=0 ttl=62 time=1.019 ms
64 bytes from 10.10.10.2: seq=1 ttl=62 time=0.723 ms
64 bytes from 10.10.10.2: seq=2 ttl=62 time=0.692 ms
64 bytes from 10.10.10.2: seq=3 ttl=62 time=2.162 ms
64 bytes from 10.10.10.2: seq=4 ttl=62 time=2.596 ms
64 bytes from 10.10.10.2: seq=5 ttl=62 time=1.867 ms
364 bytes from 10.10.10.2: seq=6 ttl=62 time=2.495 ms
64 bytes from 10.10.10.2: seq=7 ttl=62 time=0.685 ms
64 bytes from 10.10.10.2: seq=8 ttl=62 time=2.410 ms
64 bytes from 10.10.10.2: seq=9 ttl=62 time=1.180 ms
254 bytes from 10.10.10.2: seq=10 ttl=62 time=1.180 ms
64 bytes from 10.10.10.2: seq=11 ttl=62 time=0.644 ms
64 bytes from 10.10.10.2: seq=12 ttl=62 time=0.912 ms
64 bytes from 10.10.10.2: seq=13 ttl=62 time=0.498 ms
64 bytes from 10.10.10.2: seq=14 ttl=62 time=2.267 ms
64 bytes from 10.10.10.2: seq=15 ttl=62 time=0.790 ms
```

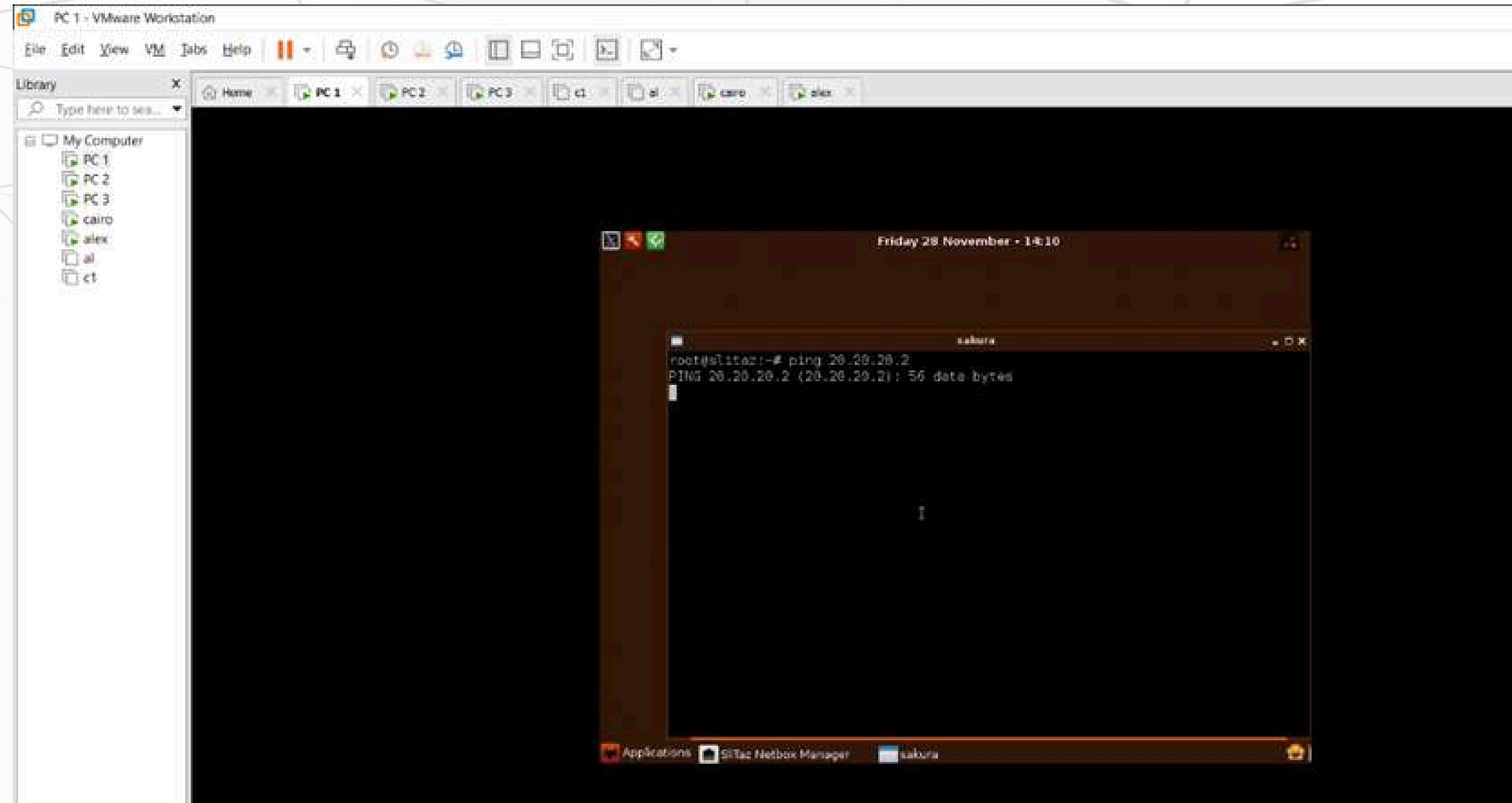
Connectivity Test Results

Comment:

- The ping was successful, confirming that the IPsec VPN tunnel is working correctly.
- Connectivity is allowed because the traffic selector includes 10.10.10.0/24 ↔ 30.30.30.0/24.

Connectivity Test Results

Ping Test: PC-1 to PC-2



Connectivity Test Results

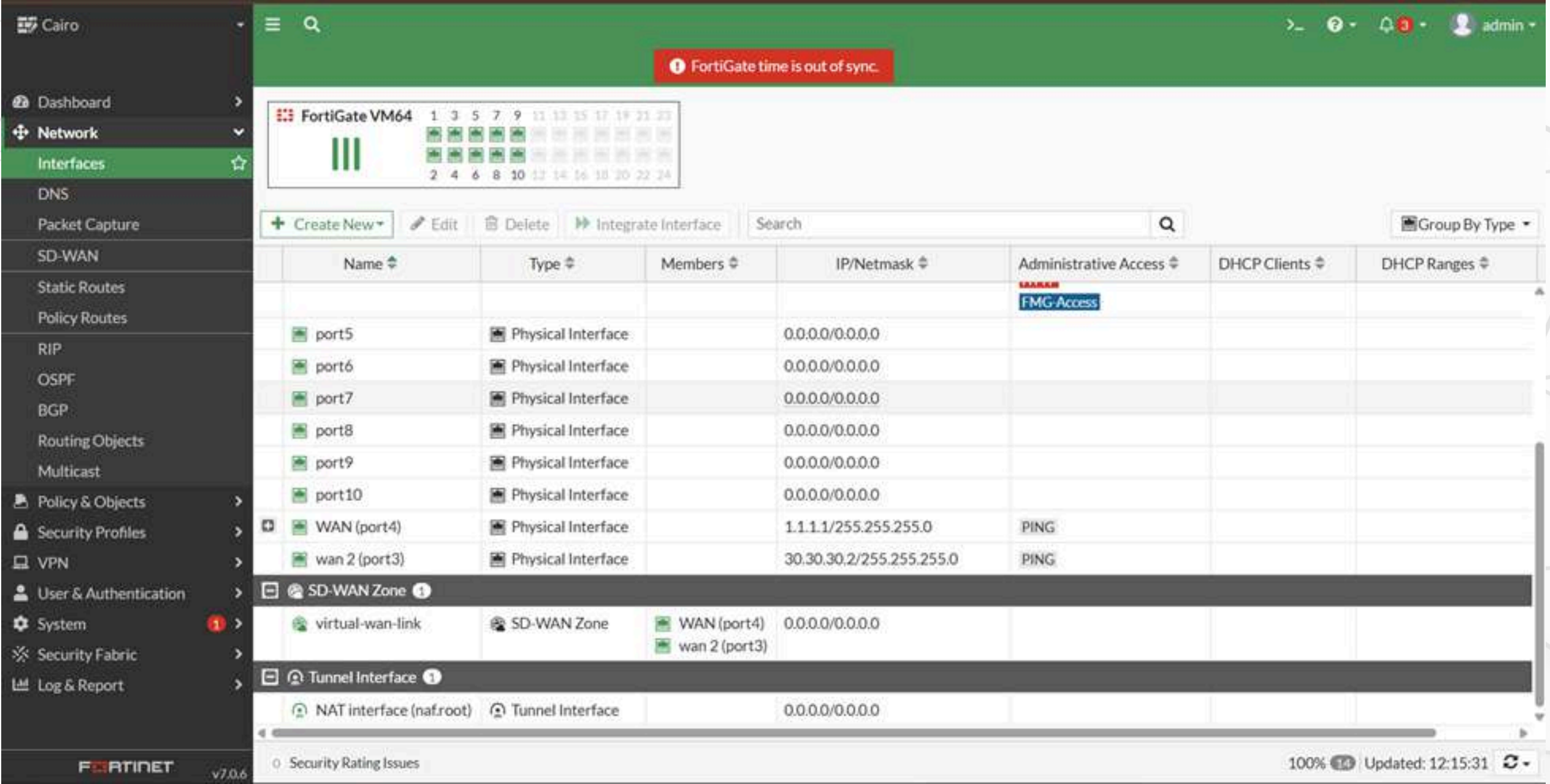
Comment:

- The ping failed, which is the expected result.
- No traffic selector is configured between the 10 network and the 20 network, so the firewall drops this traffic.

Week 3: FortiGate VPN with SD-WAN

Configuration steps of SD-WAN on FortiGate

Figure 1:configure the ports to a wan



The screenshot shows the FortiGate web interface for a device named 'Cairo'. The left sidebar contains a navigation menu with options like Dashboard, Network, Interfaces, DNS, Packet Capture, SD-WAN, Static Routes, Policy Routes, RIP, OSPF, BGP, Routing Objects, Multicast, Policy & Objects, Security Profiles, VPN, User & Authentication, System, Security Fabric, and Log & Report. The 'Network' section is expanded, and 'Interfaces' is selected. The main panel displays a table of interfaces for 'FortiGate VM64'. The table has columns for Name, Type, Members, IP/Netmask, Administrative Access, DHCP Clients, and DHCP Ranges. The interfaces listed are port5 through port10, WAN (port4), and wan 2 (port3). Below the table, there are sections for 'SD-WAN Zone' and 'Tunnel Interface'. The 'SD-WAN Zone' section shows a 'virtual-wan-link' with members 'WAN (port4)' and 'wan 2 (port3)'. The 'Tunnel Interface' section shows a 'NAT interface (naf.root)' with IP/Netmask '0.0.0.0/0.0.0.0'. A red banner at the top right indicates 'FortiGate time is out of sync.'.

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			
port8	Physical Interface		0.0.0.0/0.0.0.0			
port9	Physical Interface		0.0.0.0/0.0.0.0			
port10	Physical Interface		0.0.0.0/0.0.0.0			
WAN (port4)	Physical Interface		1.1.1.1/255.255.255.0	PING		
wan 2 (port3)	Physical Interface		30.30.30.2/255.255.255.0	PING		

SD-WAN Zone

Name	Type	Members	IP/Netmask
virtual-wan-link	SD-WAN Zone	WAN (port4) wan 2 (port3)	0.0.0.0/0.0.0.0

Tunnel Interface

Name	Type	IP/Netmask
NAT interface (naf.root)	Tunnel Interface	0.0.0.0/0.0.0.0

Security Rating Issues

100% Updated: 12:15:31

Figure 2 : create new SD-WAN member

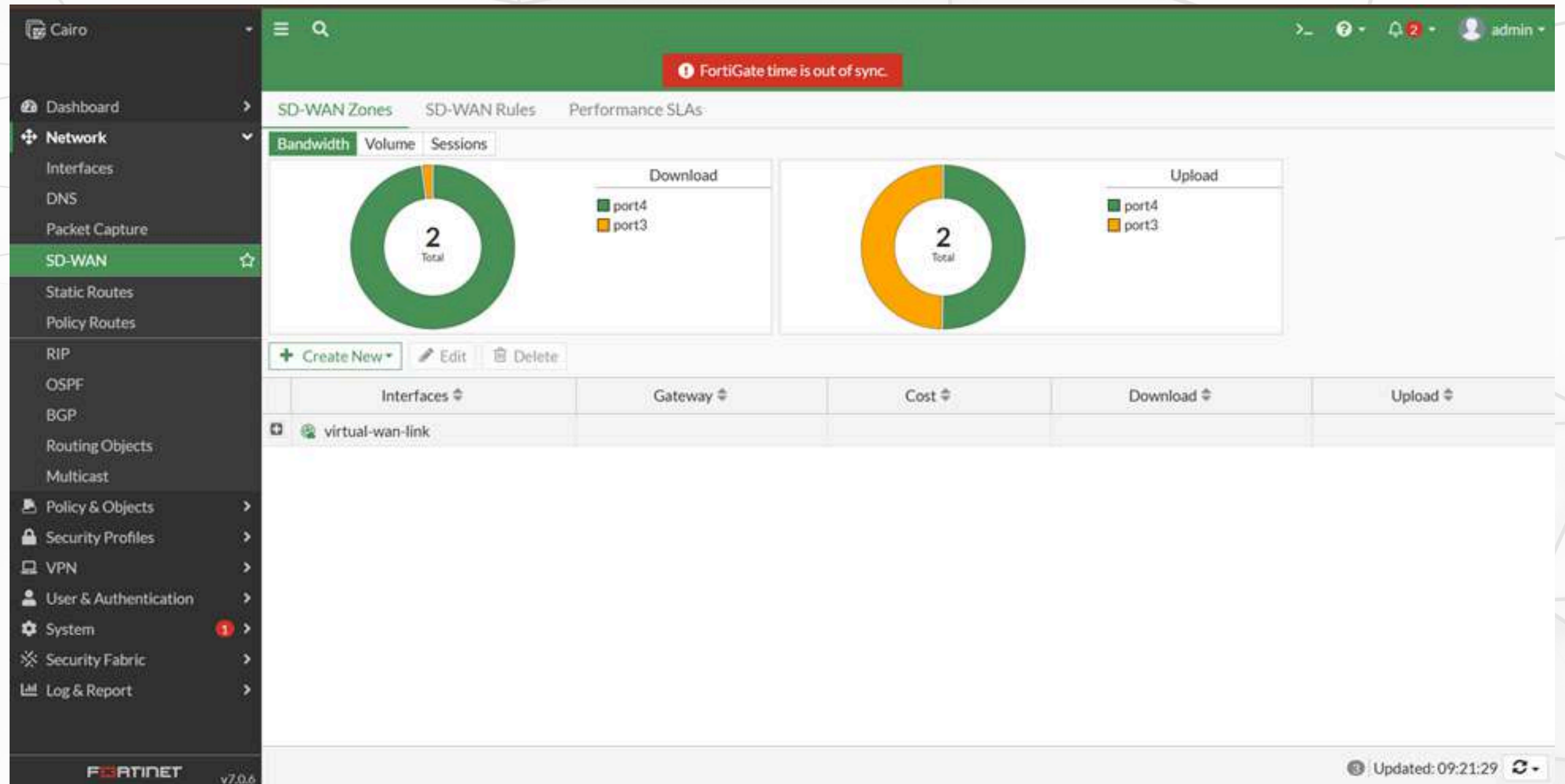
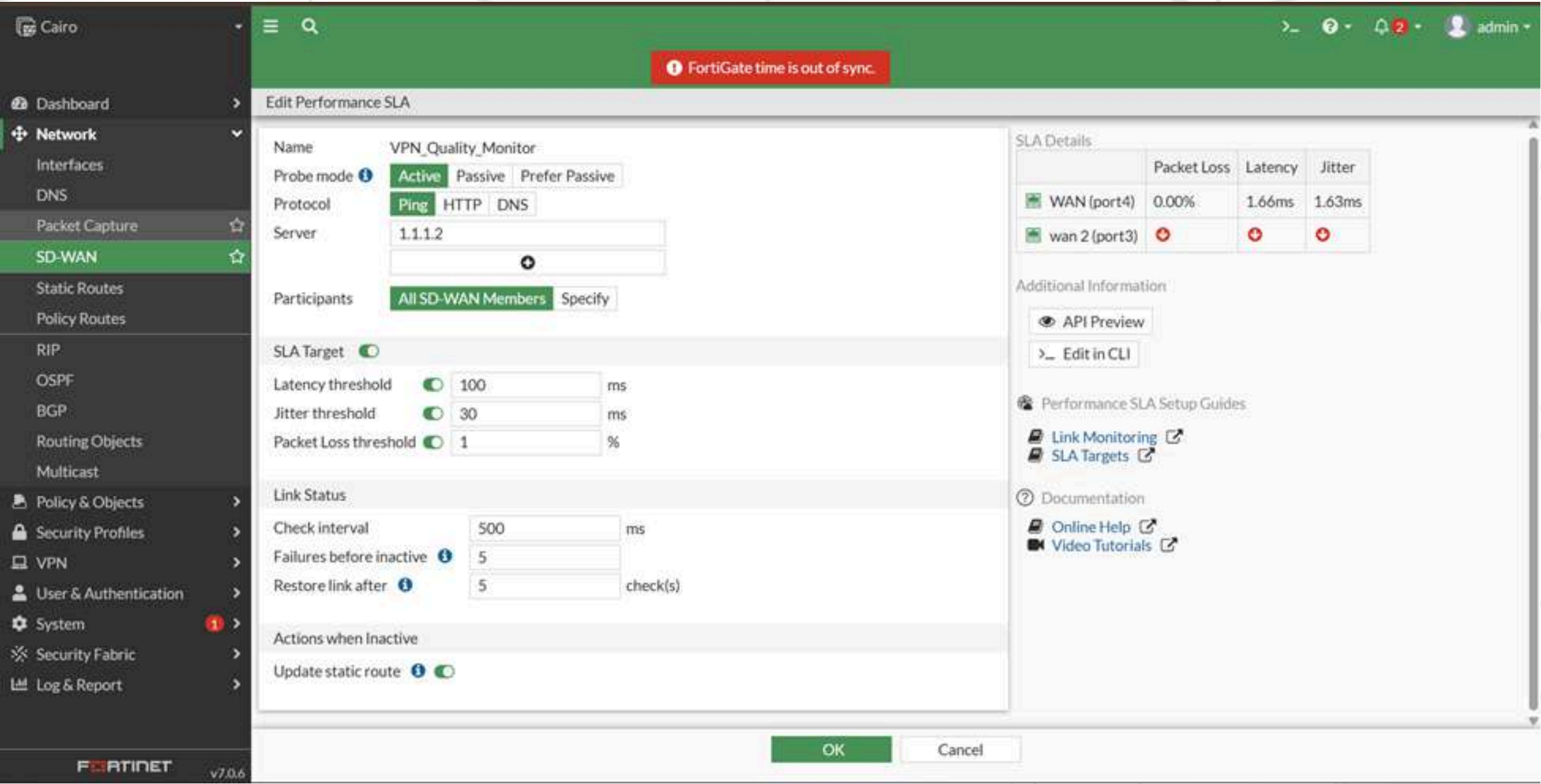


Figure 3: Define Link Performance Metrics (Performance SLA)

Purpose: This is the core intelligence. You define the acceptable quality targets (latency, jitter, loss) for the link carrying the VPN traffic. The FortiGate constantly checks both wan1 and wan2 against these targets.



The screenshot shows the FortiGate SD-WAN configuration interface for a Performance SLA. The left sidebar shows the navigation menu with 'SD-WAN' selected. The main panel is titled 'Edit Performance SLA' and contains the following configuration details:

- Name:** VPN_Quality_Monitor
- Probe mode:** Active (selected), Passive, Prefer Passive
- Protocol:** Ping (selected), HTTP, DNS
- Server:** 1.1.1.2
- Participants:** All SD-WAN Members (selected), Specify
- SLA Target:** Enabled (toggle)
- SLA Thresholds:**
 - Latency threshold: 100 ms
 - Jitter threshold: 30 ms
 - Packet Loss threshold: 1 %
- Link Status:**
 - Check interval: 500 ms
 - Failures before inactive: 5
 - Restore link after: 5 check(s)
- Actions when Inactive:**
 - Update static route: Enabled (toggle)

On the right side, there is a table for 'SLA Details' showing performance metrics for different links:

	Packet Loss	Latency	Jitter
WAN (port4)	0.00%	1.66ms	1.63ms
wan 2 (port3)	⬇	⬇	⬇

Below the table, there are links for 'API Preview', 'Edit in CLI', 'Performance SLA Setup Guides', 'Link Monitoring', 'SLA Targets', 'Documentation', 'Online Help', and 'Video Tutorials'. At the bottom right, there are 'OK' and 'Cancel' buttons.

Figure 4 : Create the SD-WAN Rule (Traffic Steering Logic)

Purpose: This rule specifically directs the VPN traffic to use the best performing link based on the real-time data collected

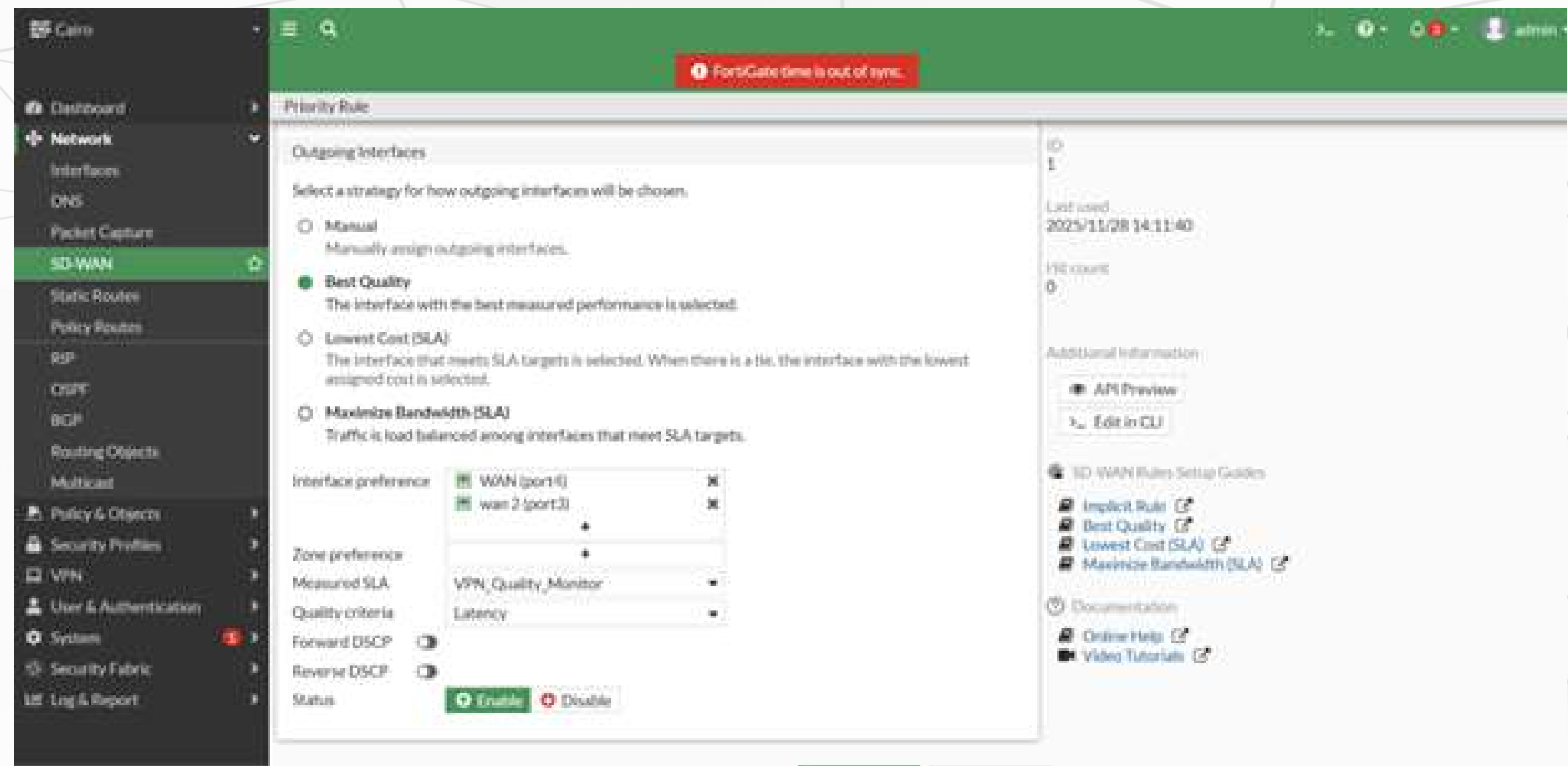
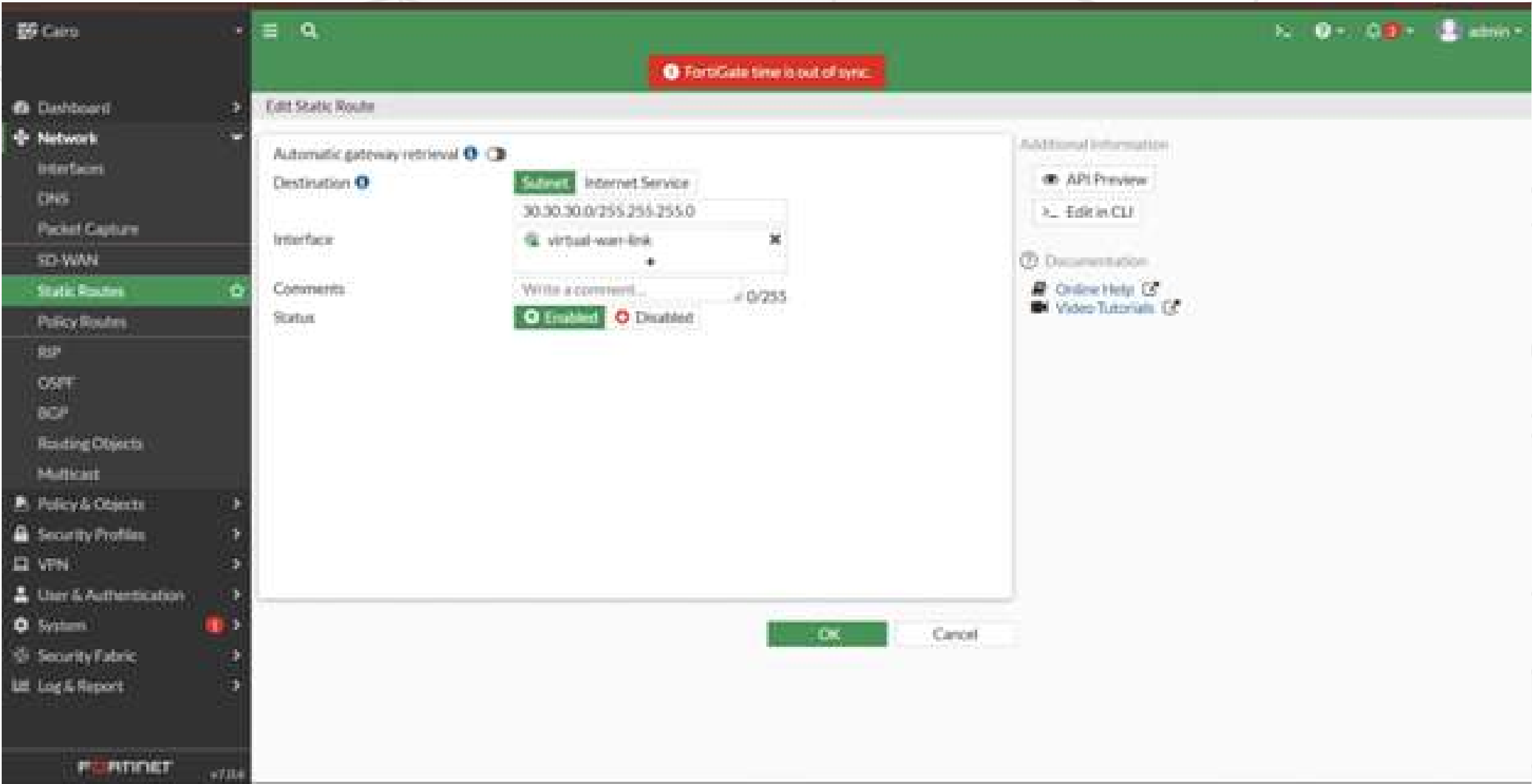


Figure 5 : Update the Static Route

Purpose: To make the SD-WAN rule effective, you must redirect the traffic bound for the remote VPN network away from a single physical interface and hand it over to the SD-WAN engine.



Conclusion

- Successfully implemented three FortiGate VPN solutions: **SSL, IPsec, and SD-WAN**.
- **SSL VPN** → Delivered secure remote access with user authentication and controlled permissions.
- **IPsec VPN** → Enabled encrypted site-to-site communication and verified tunnel stability through testing.
- **SD-WAN integration** → Improved VPN reliability by monitoring link metrics and automatically selecting optimal paths.
- Demonstrated real **configuration, troubleshooting, and validation**, not just theoretical concepts.



Any Questions ?

**THANK
YOU!**